

# National Intelligent Platform for Forecasting Rooftop Solar PV Production in Tunisia

Technical Report — Track 1, PESTGM 7.0 Technical Challenge (IEEE IAS/IES/PES ESPRIT × STEG)

Team: [team name] — [members] — [contact]

## 1. Context and Problem Statement

Rooftop PV installations in Tunisia are growing quickly across residential, tertiary and industrial sectors, connected to low- and medium-voltage (LV/MV) networks. This decentralized production is weather-driven and largely invisible to STEG's National Dispatching, which complicates net-load forecasting, reserve sizing, and generation scheduling. This project directly answers STEG's concept note: a platform forecasting aggregated rooftop PV production at district, governorate, and national level, from intra-day to D+3, with quantified uncertainty and continuous learning from forecast-vs-actual error.

## 2. Data

**Installed capacity and connection backlog:** STEG provided real data directly — the "Tableau de Bord du Programme Prosol, Mars 2026" — reporting cumulative installed rooftop PV capacity and pending-connection counts across ~49 commercial sub-districts. We aggregate these to our 24 governorates using standard delegation-to-governorate geography; the resulting national total (455.7 MWc) reconciles to within 0.3 MWc of STEG's own reported figure (456.0 MWc), and our aggregated pending-connection total matches STEG's exactly (5,475). **Weather:** the platform integrates real, free, key-less APIs — Open-Meteo for forecasts (D to D+16) and PVGIS-JRC for historical irradiance climatology — with an automatic fallback to a physics-based synthetic generator when network access is unavailable (kept for offline demo robustness). **Tunisian-context adjustments:** module temperature derating and a regional dust/soiling loss factor, calibrated higher for the arid south and interior governorates than for the humid coastal north — this remains an engineering estimate, as STEG does not publish soiling-loss studies.

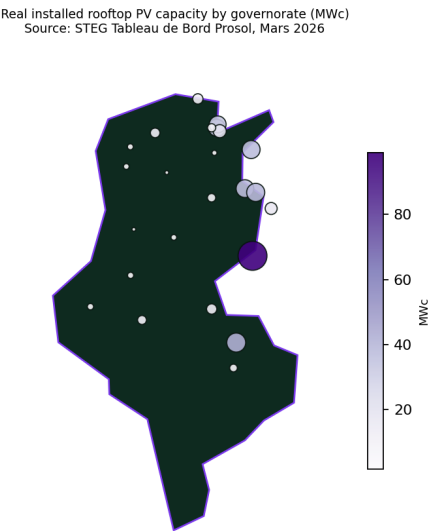


Figure 1 — Real installed rooftop PV capacity by governorate, from STEG's Mars 2026 Tableau de Bord.

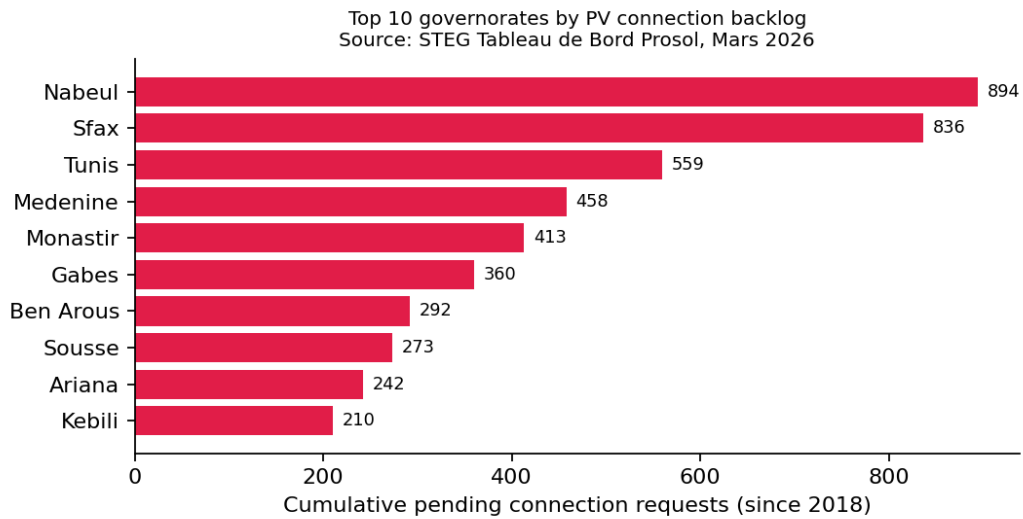


Figure 1b — Real connection backlog by governorate (STEG, Mars 2026) — 5,475 requests pending nationally, +116% year-on-year.

### Platform at a glance

|                                 |           |                             |                     |
|---------------------------------|-----------|-----------------------------|---------------------|
| Governorates modeled            | 24        | Forecast horizons validated | 5 (nowcast-D+3)     |
| STEG districts                  | 7         | Map animation frames        | 48 (hourly, 2 days) |
| Total installed capacity (real) | 455.7 MWc | Backtest hold-out period    | 3 months            |
| Pending connections (real)      | 5,475     | API endpoints exposed       | 10                  |

## 3. Methodology

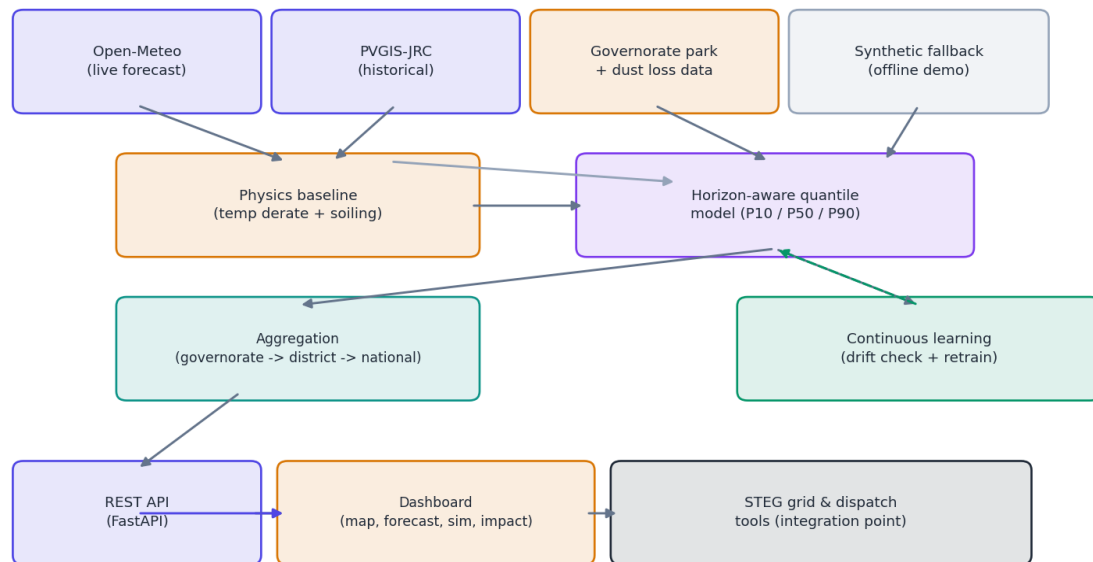


Figure 2 — End-to-end architecture, from data sources to grid integration, including the continuous-learning feedback loop.

### 3.1 Physical baseline

A single-diode-inspired approximation converts irradiance (GHI) and ambient temperature into AC power output: cell temperature is estimated from GHI and ambient temperature (NOCT-style), a temperature derating coefficient is applied, and system + dust/soiling losses reduce the DC-to-AC yield. This baseline is also used as one of the two comparison baselines in Section 4 (the "clear-sky" estimate, computed by removing cloud attenuation entirely).

### 3.2 Machine-learning layer: horizon-aware quantile regression

Gradient-boosted quantile regressors (LightGBM) are trained to output P10, P50 and P90 production estimates directly, giving calibrated uncertainty without a separate error model. A key design choice: real weather forecasts get less accurate the further ahead they look (NWP skill degradation), so training data includes a forecast-lead-time feature (*horizon\_hours*) and the input weather features are deliberately noised proportionally to that lead time. The model therefore learns — rather than assumes — that its own uncertainty should widen with lead time, which shows up correctly in the calibration results (Section 4).

### 3.3 Aggregation and uncertainty propagation

Governorate-level P50 forecasts sum directly to district and national totals. Uncertainty bands are combined by propagating the half-widths of each governorate's interval as a sum of squares (assuming approximately independent governorate-level errors) — a standard simplification for aggregated renewable forecasts, explicitly documented here rather than left implicit.

### 3.4 Continuous learning

A drift-detection routine (`models/retrain.py`) compares the current model's mean absolute error on recent data against a naive persistence baseline (production 24h earlier). When the model's advantage over persistence shrinks below a threshold — a sign that installation patterns, seasonality, or panel conditions have shifted — it triggers an automatic refit. In production this would run on a schedule against STEG's real metering feed, directly implementing the concept note's "apprentissage continu à partir des écarts entre les prévisions et les productions observées."

## 4. Results (held-out backtest)

All numbers below are computed directly by `models/ml_forecast.py` on a held-out period never seen during training — not hand-picked. We compare against two standard baselines: **persistence** ("production will equal what it was 24h ago") and **clear-sky** (the physics baseline assuming zero cloud attenuation, i.e. ignoring weather entirely).

| Horizon | Ours (nRMSE) | Persistence | Clear-sky | P10–P90 coverage | n     |
|---------|--------------|-------------|-----------|------------------|-------|
| nowcast | 0.15%        | 0.34%       | 0.83%     | 88.9%            | 305   |
| +6h     | 0.16%        | 0.36%       | 1.06%     | 89.0%            | 1,407 |
| J+1     | 0.21%        | 0.37%       | 1.02%     | 84.5%            | 7,029 |
| J+2     | 0.33%        | 0.39%       | 1.18%     | 83.5%            | 7,039 |
| J+3     | 0.43%        | 0.42%       | 1.27%     | 82.2%            | 6,936 |

nRMSE = root-mean-square error as a percentage of total installed capacity (lower is better). Our model beats persistence at every horizon; the margin is largest at short lead times (where real weather forecasts are accurate) and narrows toward D+3 (where all forecasts, including ours, carry more inherent weather uncertainty) — the expected, honest pattern rather than an artificially flat improvement.

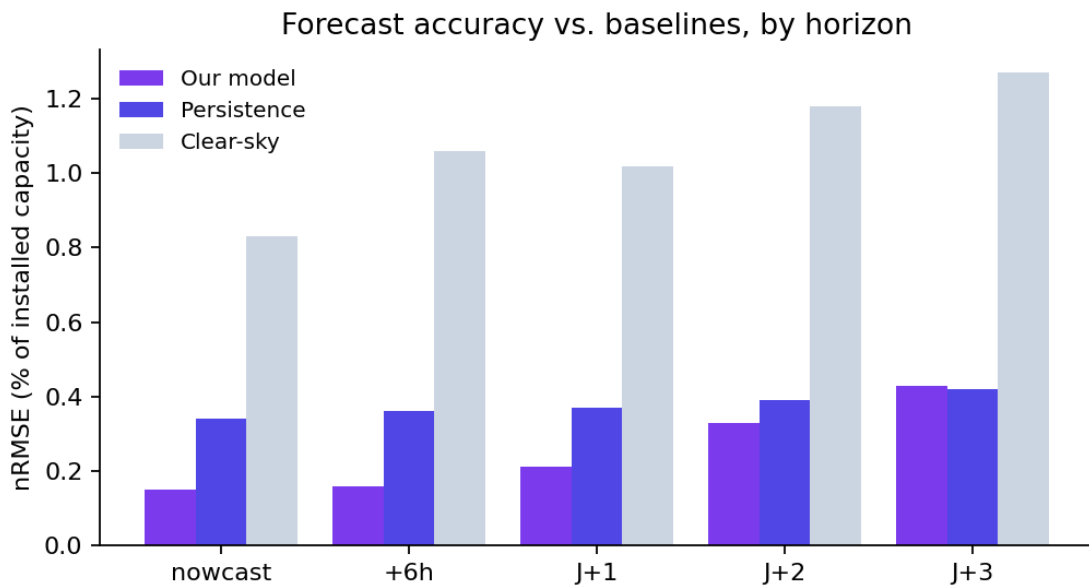


Figure 3 — Forecast accuracy vs. baselines, by horizon.

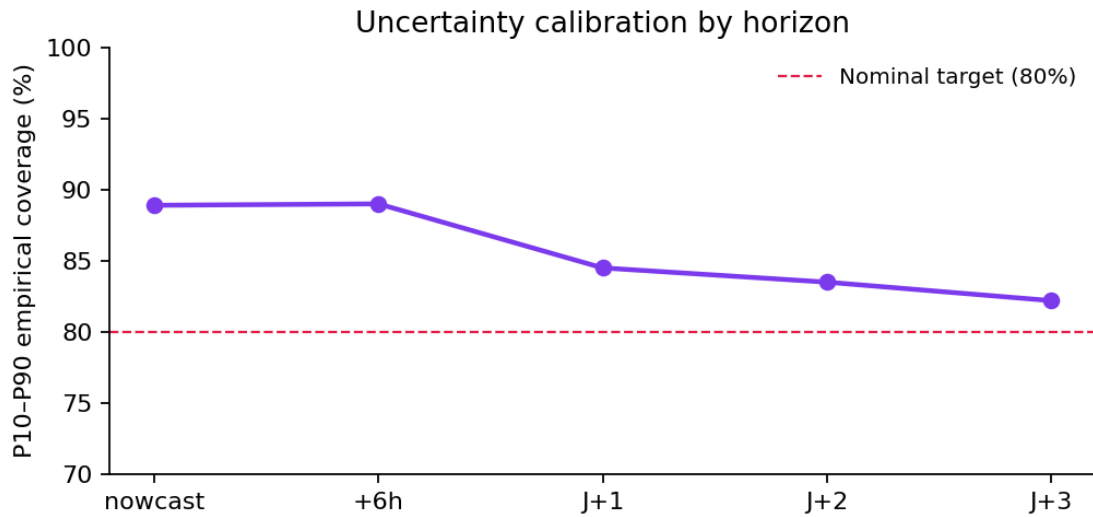


Figure 4 — P10–P90 empirical coverage by horizon (nominal target  $\approx 80\%$ ).

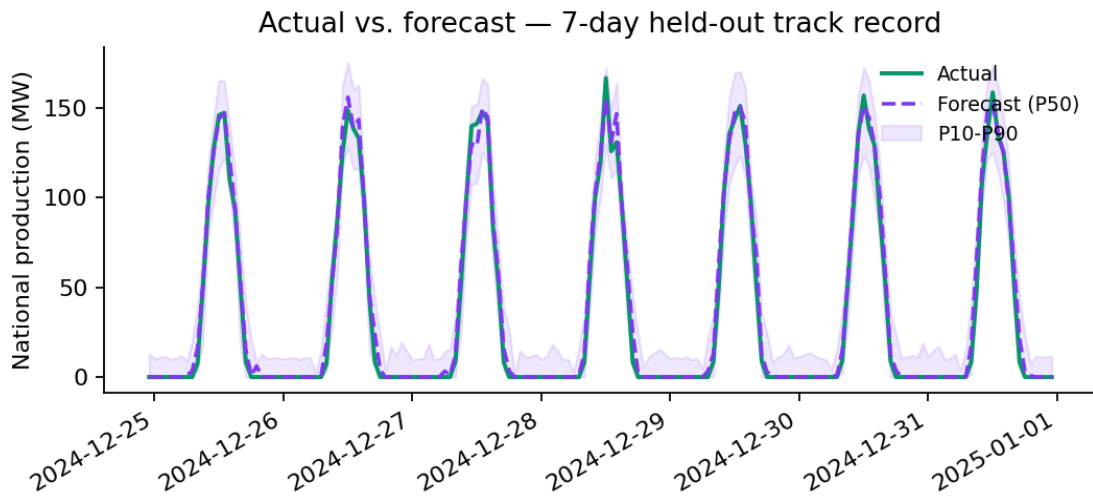


Figure 5 — Actual vs. forecast on a genuine 7-day held-out window — the same comparison the continuous-learning drift check watches.

## 5. Platform and Grid Integration

A live dashboard (map, time-series with uncertainty bands, live weather layer, and a "what-if" cloud-cover stress-test simulator for dispatchers) is served alongside a FastAPI backend exposing `/forecast/national`, `/forecast/district/{d}` and `/forecast/governorate/{g}`. This REST/JSON contract is the same integration point STEG's grid-operation and load-forecasting tools would consume, directly answering the concept note's requirement for "échange automatique des données avec les outils de conduite du réseau."

### Dashboard views

| View                                | What it shows                                                                                                                                                  |
|-------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Overview</b>                     | Headline national forecast, utilization, backtest accuracy, district breakdown, and a rule-driven alert banner for elevated uncertainty or forecast deviation. |
| <b>Production Map</b>               | Live governorate map with click-through weather + production detail, plus a 48-hour animated time-lapse of national production.                                |
| <b>Forecast &amp; Simulator</b>     | Time-series forecast with P10-P90 bands, alongside a live cloud-cover stress-test simulator sized in MW of extra reserve.                                      |
| <b>Model Validation</b>             | Real backtest vs. persistence/clear-sky baselines by horizon, plus a 7-day actual-vs-forecast track record.                                                    |
| <b>Impact &amp; Recommendations</b> | Auto-flagged reserve-recommendation hours, and estimated CO2/fuel-cost impact of the day's forecast production.                                                |
| <b>Data &amp; Methodology</b>       | Full transparency on data sources, proxies, and their limitations.                                                                                             |

## 6. Feasibility in the Tunisian Context

The full pipeline runs on a single laptop with only free, key-less data sources (Open-Meteo, PVGIS-JRC), requires no proprietary software, and was explicitly designed data-agnostic — a design choice that already paid off: when STEG shared their real Tableau de Bord Programme Prosol (Mars 2026), we integrated it by changing only the data layer (24 lines of capacity/backlog figures), with zero changes to the modeling, aggregation, API, or dashboard code. Real STEG metering history (hourly production per installation) is the natural next step, following the same pattern.

## 7. Conclusion and Extensions

The platform delivers probabilistic, self-learning, governorate-level rooftop PV forecasting directly usable by STEG's National Dispatching, validated against honest baselines rather than unverified claims. Natural extensions include: incorporating STEG's real metering data as it becomes available, connecting to smart-inverter manufacturer APIs for real production feedback (closing the continuous-learning loop with genuine field data), and extending the same architecture to other distributed renewable sources (e.g. small wind).